AI511/MM505 Linear Algebra with Applications

Lecture 11 – Determinants (continued), Eigenvalues and Eigenvectors

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Outline

Recap

Determinants (continued)

Similar matrices

Eigenvalues and eigenvectors

- Definition and computation
- Characteristic polynomial and algebraic multiplicity

Recap

Rank and nullity

Definition 2.4.4 of Johnston (2021)

The *rank* of $A \in \mathbb{R}^{m \times n}$ is denoted by $\textsf{rank}(A)$ and defined by

$$\textsf{rank}(A) = \textsf{dim}(\textsf{range}(A)).$$

The rank of a matrix can be thought of as a measure of how non-invertible a square matrix is (i.e., how small the subspace to which it squashes $\mathbb{R}^n$ is).

Definition

The *nullity* of $A \in \mathbb{R}^{m \times n}$ is denoted by $\textsf{nullity}(A)$ and defined by

$$\textsf{nullity}(A) = \textsf{dim}(\textsf{null}(A)).$$

Invertibility of a matrix

Theorem 2.4.9 of Johnston (2021)

Let $A \in \mathbb{R}^{n \times n}$. The following are equivalent:

(a) A is invertible;

(b) $\text{rank}(A) = n$;

(c) $\text{nullity}(A) = 0$.

Determinant

- The determinant of A, which we denote by $\det(A)$ or $|A|$, is the area (or volume, or hypervolume, depending on the dimension n) of this distortion of the unit square.
- In other words, it measures how much A expands space when acting as a linear transformation—it is the ratio

$$\frac{\text{volume of output region}}{\text{volume of input region}}.$$

Determinants (continued)

Diagonal matrices

Definition

$A \in \mathbb{R}^{n \times n}$ is said to be *diagonal* if $a_{ij} = 0$ for $i \neq j$.

Examples

$$\begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

If $A \in \mathbb{R}^{n \times n}$ is diagonal, we also write

$$A = \mathsf{diag}(\ a_1 \quad \ldots \quad a_n\ ).$$

Diagonal matrices (continued)

Matrix multiplication of diagonal matrices is performed elementwise

$$\mathbf{diag}(\ a_1 \quad \dots \quad a_n\ )\,\mathbf{diag}(\ b_1 \quad \dots \quad b_n\ ) = \\ = \mathbf{diag}(\ a_1 b_1 \quad \dots \quad a_n b_n\ ).$$

Theorem

A diagonal matrix $D \in \mathbb{R}^{n\times n}$ is invertible if and only if $d_i \neq 0$ for all $i = 1, \dots, n$ and in which case $D^{-1} = \mathbf{diag}(\ d_1^{-1} \quad \dots \quad d_n^{-1}\ )$.

It is straightforward to calculate powers of a diagonal matrix D since

$$D^k = \mathbf{diag}(\ d_1^k \quad \dots \quad d_n^k\ )$$

for $k \geq 1$.

Triangular matrices

Definition

$A \in \mathbb{R}^{n \times n}$ is called *upper (lower) triangular* matrix if all entries below (above) the main diagonal are zero, i.e., $a_{ij} = 0$ for $i > j$ ($a_{ij} = 0$ for $i < j$).

Examples

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

Upper triangular

and

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}.$$

Lower triangular

So diagonal means upper *and* lower triangular.

Determinants of triangular matrices

Theorem 3.2.4 of Johnston (2021)

Suppose $A \in \mathbb{R}^{n \times n}$ is a triangular matrix. Then $\det(A)$ is the product of its diagonal entries:

$$\det(A) = a_{11} \cdot \ldots \cdot a_{nn}.$$

Example

$$\begin{vmatrix} 1 & 2 & -1 & 7 & 91 \\ 0 & 1 & 19 & 0 & \frac{1}{10} \\ 0 & 0 & 2 & 1 & -5 \\ 0 & 0 & 0 & 1 & \pi \\ 0 & 0 & 0 & 0 & 3 \end{vmatrix} = 1 \cdot 1 \cdot 2 \cdot 1 \cdot 3 = 6.$$

Recall that a diagonal matrix is also triangular.

Determinant of 2×2 matrices

Theorem 3.2.6 of Johnston (2021)

The determinant of a 2×2 matrix is given by

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc.$$

We saw this expression when we talked about the inverse of a 2×2 matrix and we saw that a 2×2 matrix is invertible if and only if $ad - bc \neq 0$.

Minors and cofactors

Definition 3.2.2 of Johnston (2021)

Suppose $A \in \mathbb{R}^{n \times n}$ and let $1 \leq i, j \leq n$. Then

(a) the (i, j)-minor of A is the determinant of the $(n-1) \times (n-1)$ matrix that is obtained by deleting the i-th row and j-th column of A, and

(b) the (i, j)-cofactor of A is the quantity $(-1)^{i+j} m_{ij}$, where m_{ij} is the (i, j)-minor of A.

Computing determinants via cofactor expansion

The determinants of large matrices can be expressed in terms of determinants of smaller matrices.

Theorem 3.2.8 of Johnston (2021)

Suppose $A \in \mathbb{R}^{n \times n}$ has cofactors $\{c_{ij}\}_{i,j=1}^{n}$. Then

$$\det(A) = a_{i1}c_{i1} + a_{i2}c_{i2} + \ldots + a_{in}c_{in} \tag{1}$$

$$\det(A) = a_{1j}c_{1j} + a_{2j}c_{2j} + \ldots + a_{nj}c_{nj} \tag{2}$$

for all $i, j = 1, \ldots, n$.

- (1) is called a cofactor expansion along the i-th row.
- (2) is called a cofactor expansion along the j-th column.

Examples

- We have that

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a(-1)^{1+1}d + b(-1)^{1+2}c = ad - bc.$$

- Suppose that

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}.$$

  Show that $\det(A) = -1$.

Tips and tricks

- Instead of row operations, one can do column operations. This corresponds to row operations on A^T.
- If a matrix A has a row or column with a lot of zeros, then this often good for row/column cofactor expansion.
- If a matrix A has a zero row (or column) then $\det(A) = 0$.
- If two rows (columns) are proportional, then $\det(A) = 0$, since one can then create a zero-row (zero-column) by row operations.

Similar matrices

Similar matrices

Definition 3.1.3 of Johnston (2021)

We say that two matrices $A, D \in \mathbb{R}^{n \times n}$ are similar if there exists an invertible $P \in \mathbb{R}^{n \times n}$ such that $A = PDP^{-1}$.

- D is an arbitrary matrix in the definition (not necessarily diagonal).
- For example,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 5 & -1 \\ -2 & 0 \end{bmatrix}$$

  are similar with

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

What does it mean that two matrices are similar?

- Let $p_1, \ldots, p_n \in \mathbb{R}^n$ be the columns of the invertible matrix P and
$$B = \{p_1, \ldots, p_n\}$$
be a basis of $\mathbb{R}^n$ (think why this is the case).
- It follows that
$$P = P_{C \leftarrow B} \quad \text{and} \quad P^{-1} = P_{B \leftarrow C},$$
where C is the standard basis of $\mathbb{R}^n$.
- The expression
$$A = PDP^{-1}$$
shows that the transformation A in the standard basis is the same as the transformation D in the basis B.
- So what? The transformation D might be much simpler (diagonal) than the transformation A.

Motivating example

- Suppose we want to compute A^k, where $A \in \mathbb{R}^{n \times n}$ and k is a very large integer.
- If A is a general matrix (i.e., no special structure), computing A^k requires a lot of matrix multiplications and hence is computationally intensive.
- If A is a diagonal matrix, computing A^k is fairly straightforward since

$$A^k = \mathbf{diag}(a_{11}^k, a_{22}^k, \ldots, a_{nn}^k).$$

Motivating example (cont.)

- Suppose that A is not diagonal but it is similar to some diagonal matrix D, i.e.,
$$A = PDP^{-1}.$$
- Then we have that
$$A^k = \underbrace{(PD\overbrace{P^{-1}}^{=I})(P\,D\overbrace{P^{-1}}^{=I})(P\,DP^{-1})\dots(PDP^{-1})}_{k \text{ times}} = PD^kP^{-1}.$$
- Since D is diagonal, calculating D^k is straightforward and there are only left two matrix multiplications: one on the left by P and one on the right by P^{-1}.
- It costs to obtain P, D, and P^{-1} but once we have these three matrices, we can do a lot of operations in a much cheaper way.

Questions

- Are all matrices similar to diagonal matrices?
- If a matrix is similar to a diagonal matrix, how do we find P and D?

Eigenvalues and eigenvectors

Eigenvalues and eigenvectors

Definition and computation

Eigenvalues and eigenvectors

Sometimes matrix multiplication behaves like scalar multiplication.

Definition 3.3.1 of Johnston (2021)

Suppose A is a square matrix. A non-zero vector v is called an *eigenvector* of A if there is a scalar λ such that

$$Av = \lambda v.$$

Such a scalar λ is called the *eigenvalue* of A corresponding to v.

- If v is an eigenvector, so is cv with $c \neq 0$. Pay attention that 0 is not an eigenvector according to the definition.
- Recommended to watch *Eigenvectors and eigenvalues* of 3Blue1Brown.

Diagonal matrices

If $A = \mathbf{diag}(a_{11}, \ldots, a_{nn})$, then $e_1, \ldots, e_n$ are the eigenvectors of A and $a_{11}, \ldots, a_{nn}$ are the corresponding eigenvalues since

$$Ae_i = a_{ii}e_i$$

for $i = 1, \ldots, n$.

A two-step procedure: step 1

- λ is an eigenvalue of A if and only if there exists a non-zero vector v such that
$$Av = \lambda v.$$
- We have that
$$Av = \lambda v \quad \iff \quad Av - \lambda v = 0 \quad \iff \quad (A - \lambda I)v = 0.$$
- When does a homogeneous systems of linear equations have a non-zero solution?
- A scalar λ is an eigenvalue of A if and only if $A - \lambda I$ is not invertible.
- $A - \lambda I$ is not invertible if and only if $\det(A - \lambda I) = 0$.

Blackboard example: calculating eigenvalues

Compute the eigenvalues of

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}.$$

A two-step procedure: step 2

- Once we know the eigenvalues of a matrix, the eigenvectors can be found by solving the linear system $Av = \lambda v$ for v.
- This is equivalent to solving the system of linear equations

$$(A - \lambda I)v = 0$$

  (i.e., computing the null space of the matrix $A - \lambda I$).
- A non-zero vector v is an eigenvector of A with corresponding eigenvalue λ if and only if $v \in \mathbf{null}(A - \lambda I)$.

Blackboard example (cont.)

Compute the eigenvectors of

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

using the eigenvalues $\lambda_1 = -1$ and $\lambda_2 = 6$ from the earlier computations.

Eigenvalues and eigenvectors

Characteristic polynomial and algebraic multiplicity

The characteristic polynomial

$\det(A - \lambda I)$ is a polynomial in λ which is set to 0 and solved for λ when eigenvalues are computed.

Definition 3.3.2 of Johnston (2021)

Suppose A is a square matrix. The function $p_A : \mathbb{R} \to \mathbb{R}$ defined by

$$p_A(\lambda) = \det(A - \lambda I)$$

is called the *characteristic polynomial* of A.

For example, the characteristic polynomial of

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

is $p_A(\lambda) = \lambda^2 - 5\lambda - 6$.

Properties of the characteristical polynomial

- The characteristic polynomial of an $n \times n$ matrix has degree n.
- Every degree-n polynomial has at most n distinct roots.
- Every $n \times n$ matrix has at most n distinct eigenvalues.

Definition 3.3.3 of Johnston (2021)

Suppose that $A \in \mathbb{R}^{n \times n}$ with eigenvalue λ. The *algebraic multiplicity* of λ is its multiplicity as a root of A's characteristic polynomial.

- Algebraic multiplicity tells us "how many times" the particular eigenvalue occurs as a root of the characteristic polynomial.
- Roots with multiplicity 2 or greater are sometimes called "repeated roots".

Blackboard example: eigenvalues and algebraic multiplicities

Compute the eigenvalues, and their algebraic multiplicities, of the matrix

$$A = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

References



Johnston, Nathaniel (2021). ***Introduction to linear and matrix algebra.*** Springer Cham.